

LAB EXPERIMENT RECORD

Set 11: Product Category Analysis

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Aim

To analyze product category sales using a funnel chart, tabular representation, Tableau dashboard, and pie chart, and to visualize the distribution of sales across product categories.

Dataset: Product Categories

Category	Sales (in \$)
Electronics	50000
Clothing	35000
Appliances	40000

PROGRAM 1: Funnel Chart for Product Category Sales

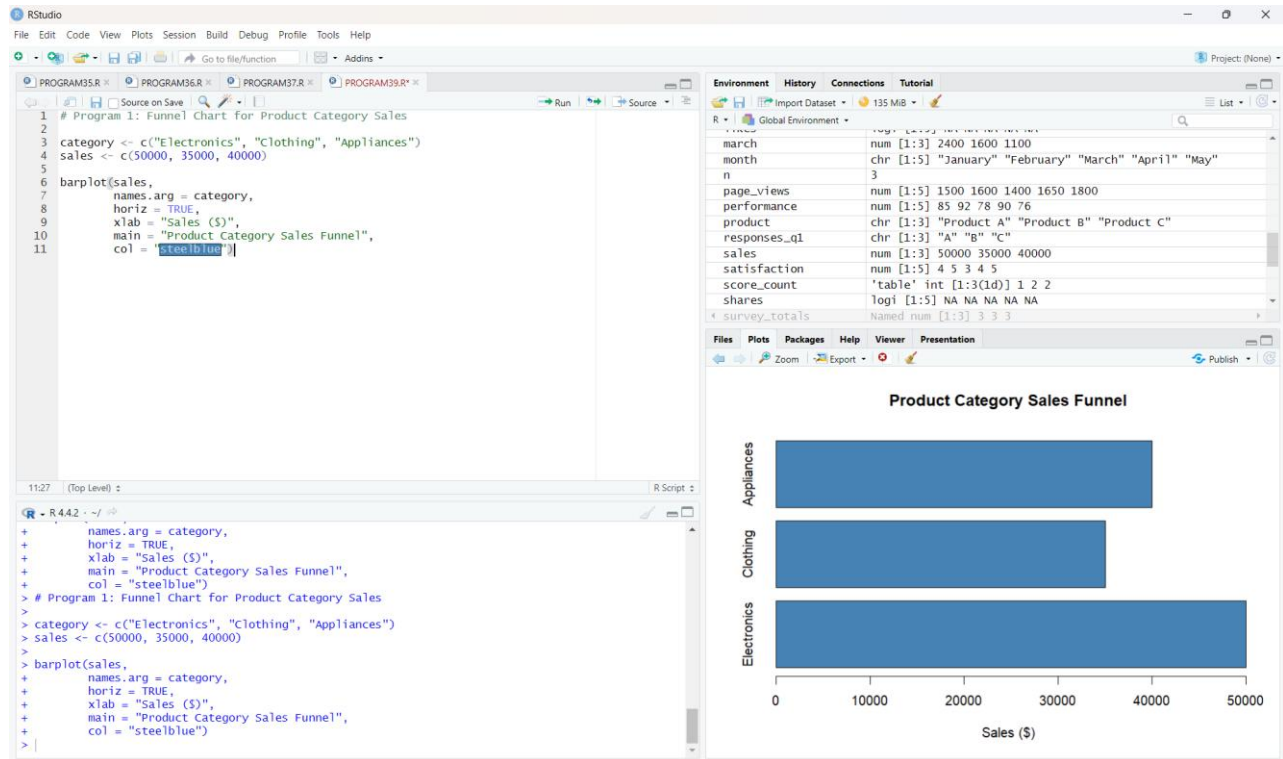
Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.

```
# Program 1: Funnel Chart for Product Category Sales
```

```
category <- c("Electronics", "Clothing", "Appliances")
sales <- c(50000, 35000, 40000)
```

```
barplot(sales,
        names.arg = category,
        horiz = TRUE,
        xlab = "Sales ($)",
        main = "Product Category Sales Funnel",
        col = "steelblue")
```

Output 1: Product Category Sales Funnel



Output 1: Funnel Chart for Product Category Sales

Observation: The funnel-style horizontal bar chart compares sales across the three product categories. Electronics has the highest sales at \$50,000, followed by Appliances at \$40,000 and Clothing at \$35,000.

PROGRAM 2: Product Category Sales Table

Build a table to display the sales data for each product category. Label the table elements.

```
# Program 2: Product Category Sales Table
```

```
product_data <- data.frame(  
  Category = c("Electronics", "Clothing", "Appliances"),  
  Sales = c(50000, 35000, 40000)  
)  
  
print(product_data)
```

Output 2: Product Category Sales Table

The screenshot shows the RStudio interface. The script editor on the left contains the following code:

```
1 # Program 2: Product Category Sales Table
2
3 product_data <- data.frame(
4   Category = c("Electronics", "Clothing", "Appliances"),
5   Sales = c(50000, 35000, 40000)
6 )
7
8 print(product_data)
```

The Environment pane on the right shows the objects created:

- `product_data`: 3 obs. of 2 variables
- `sales_data`: 3 obs. of 4 variables
- `survey_data`: 3 obs. of 4 variables
- `ui`: List of 4

The console at the bottom shows the output of the `print` function:

```
> print(product_data)
  Category Sales
1 Electronics 50000
2  Clothing  35000
3  Appliances 40000
```

Output 2: Product Category Sales Table

Observation: The table clearly displays each product category and its corresponding sales value.

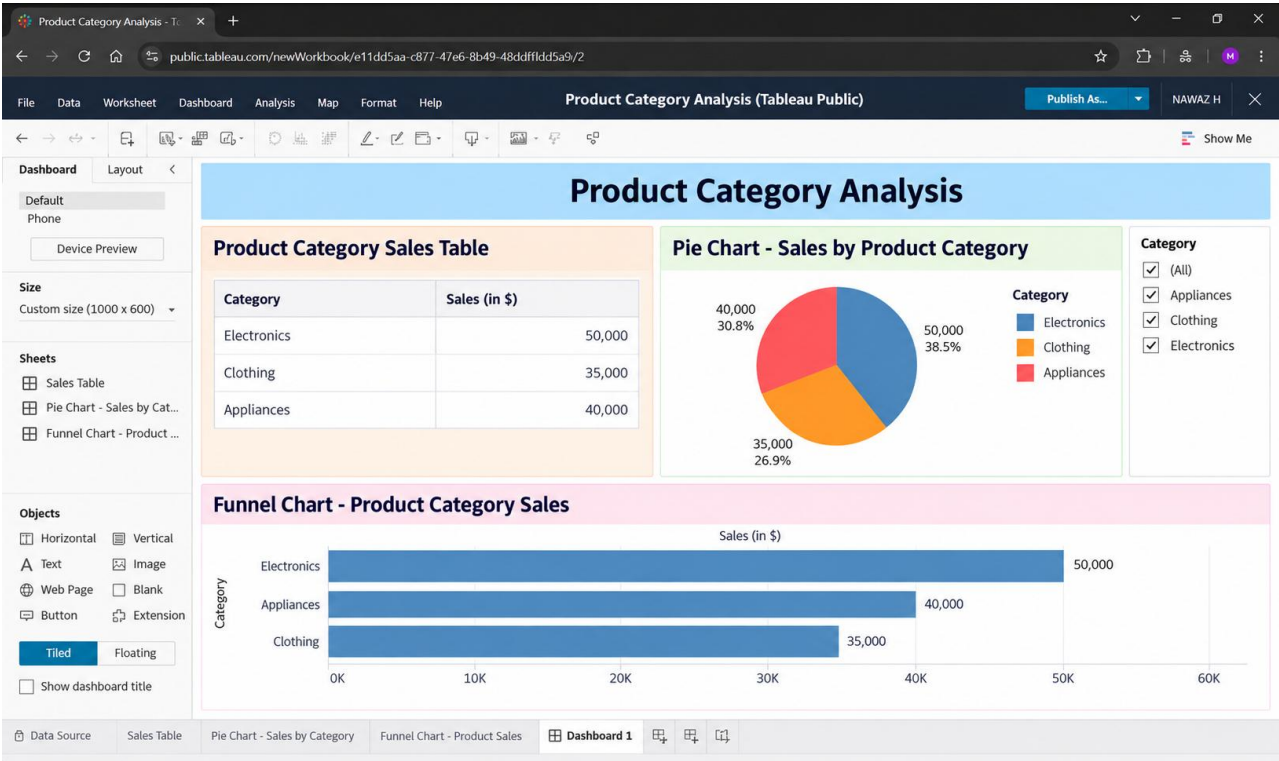
PROGRAM 3: Tableau Dashboard

Develop a Tableau dashboard combining the pie chart, funnel chart, and table for interactive exploration of product category data.

TABLEAU DASHBOARD STEPS

1. Enter the Product Categories dataset into Tableau Public.
2. Create a Sales Table worksheet using Category and Sales.
3. Create a Pie Chart using Category and Sales.
4. Create a Funnel Chart using Category and Sales as a horizontal bar chart.
5. Create a new dashboard and place the Sales Table, Pie Chart, and Funnel Chart together.
6. Add the Category filter for interactive exploration.
7. Set the dashboard title to "Product Category Analysis".

Output 3: Tableau Product Category Analysis Dashboard



Output 3: Tableau Product Category Analysis Dashboard

Observation: The Tableau dashboard combines the sales table, pie chart, and funnel chart in one interactive view. The Category filter allows individual product categories to be explored.

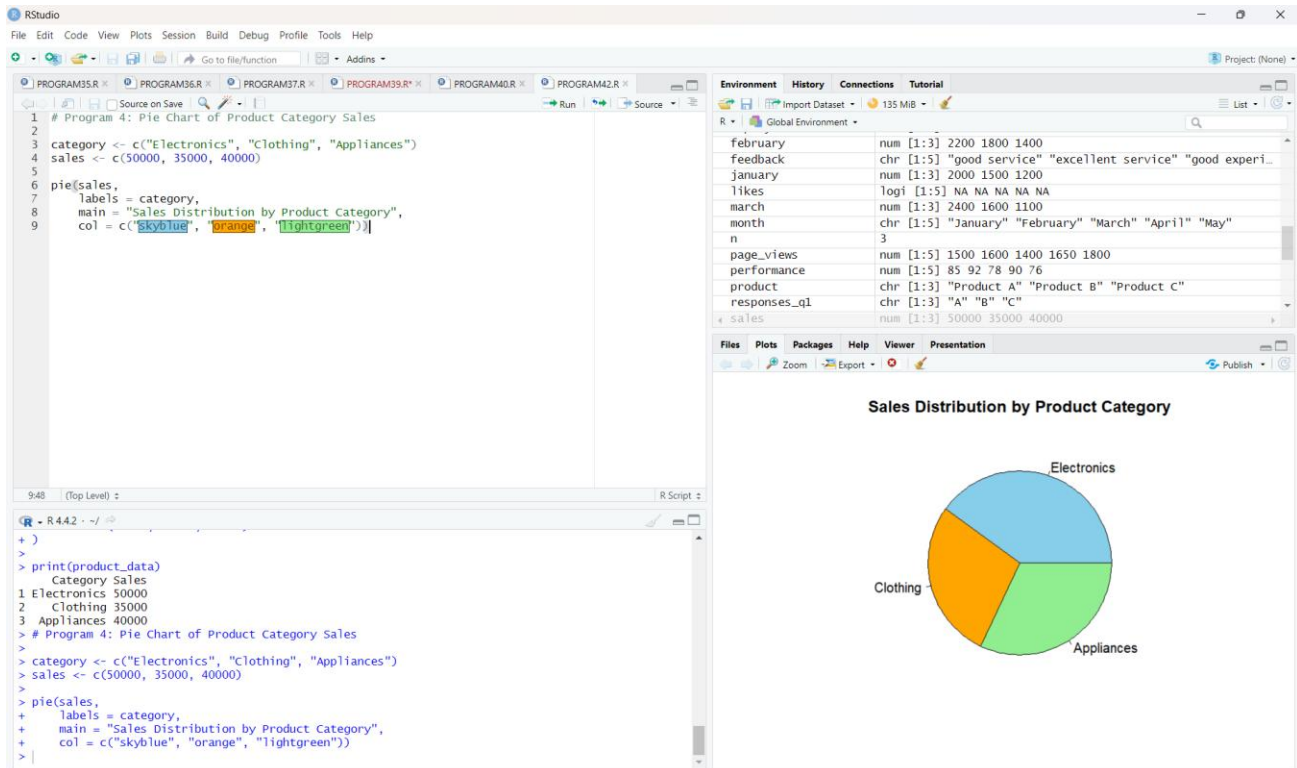
PROGRAM 4: Pie Chart of Product Category Sales

Create a pie chart to represent the distribution of sales across product categories. Include labels.

Program 4: Pie Chart of Product Category Sales

```
category <- c("Electronics", "Clothing", "Appliances")
sales <- c(50000, 35000, 40000)
```

```
pie(sales,
    labels = category,
    main = "Sales Distribution by Product Category",
    col = c("skyblue", "orange", "lightgreen"))
```



Output 4: Sales Distribution by Product Category

Observation: The pie chart shows the share of total sales contributed by each category. Electronics contributes 40.0%, Appliances 32.0%, and Clothing 28.0%.

Procedure

1. Create the product category and sales vectors in R.
2. Use `barplot()` with horizontal orientation to create the funnel-style product category sales chart.
3. Create and print the product category sales table using `data.frame()`.
4. Enter the product category sales data into Tableau and create the Sales Table, Pie Chart, and Funnel Chart worksheets.
5. Combine the Tableau worksheets into a dashboard, add the Category filter, and format the dashboard for interactive exploration.
6. Create the pie chart in R using `pie()` and include category labels.

Result

The Product Category Analysis was successfully completed using R and Tableau. The sales data was represented through a funnel chart, table, interactive Tableau dashboard, and pie chart.

Key R Functions Used

Function	Purpose
<code>barplot()</code>	Creates the horizontal funnel-style sales chart.
<code>data.frame()</code>	Creates the product category sales table.
<code>pie()</code>	Creates the pie chart for sales distribution.
<code>print()</code>	Displays the product category sales data.